

# Rigid Local Systems

## Candidacy Presentation

[Your name]

[Department]  
[University]

[Presentation date]

# Roadmap

- 1 Motivation
- 2 Background and definitions
- 3 Main result
- 4 Examples and next steps

# The guiding question

## Question

When does local monodromy determine a local system up to isomorphism?

*[Explain the motivation and give one concrete example.]*

Let

$$U = \mathbb{P}^1(\mathbb{C}) \setminus S, \quad S = \{s_1, \dots, s_m\}.$$

- *[Specify the coefficient field and rank.]*
- *[Define the local systems under consideration.]*
- *[Explain the local monodromy at each puncture.]*

## Definition (Physical rigidity)

*[Insert the precise definition, including the class of competing local systems and the meaning of the same local monodromy.]*

*[Distinguish local conjugacy from a single global conjugacy.]*

# The main theorem

Theorem (*[Attribution and theorem number]*)

*[State every hypothesis and the exact conclusion.]*

*[Explain the significance in one sentence.]*

# Proof strategy

- ① *[First reduction or key construction.]*
- ② *[Central calculation or lemma.]*
- ③ *[Deduce the theorem.]*

# A key step

## Lemma

*[State the key lemma.]*

## Proof idea.

*[Give the main argument; move lengthy details to the appendix.]*



## A worked example

*[Choose a low-rank example and compute its local data.]*

$$A_1 \cdots A_m = I, \quad A_i \in \mathrm{GL}_n(\mathbb{C}).$$

*[Explain what the theorem tells us in this example.]*

## Next steps

- *[A mathematical question to investigate.]*
- *[A formalization task, with precise remaining dependencies.]*
- *[A concrete milestone.]*

# References

*[Add the references used in the talk.]*

## Backup: details and computations

*[Put additional proofs, examples, or definitions here.]*